Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If the momentum of electron is changed by P, then the de Broglie wavelength associated with it changes by 0.5%. The initial momentum of electron will be :

(A) $200P$

(B) $400P$

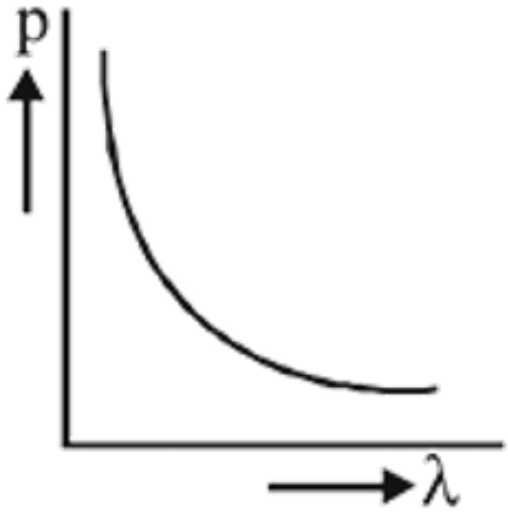
(C) $\frac{P}{200}$

(D) $100P$

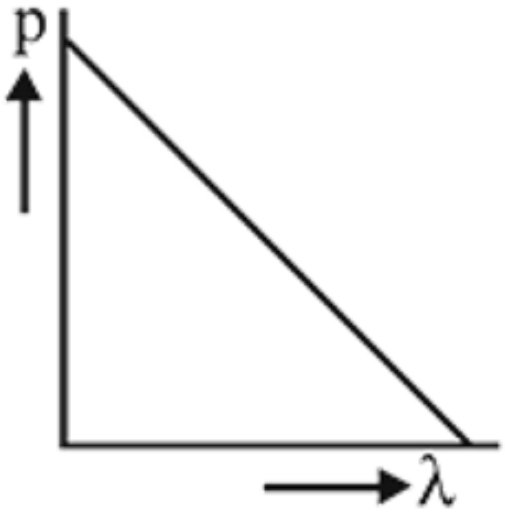
Q.2

Which of the following figures represents the variation of particle momentum and the associated de-Broglie wavelength?

(A)

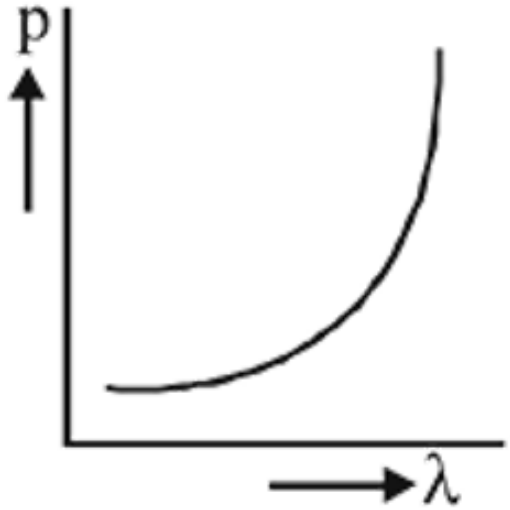


(B)

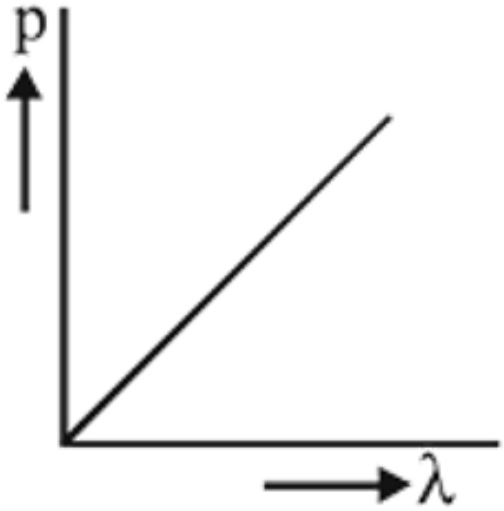


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(C)



(D)



Q.3

Electrons are accelerated through a potential difference V and protons are accelerated through a potential difference $4\ \mathrm{V}$. The de-Broglie wavelengths are λ_e and λ_p for electrons and protons respectively. The ratio of $\frac{\lambda_e}{\lambda_p}$ is given by: (given m_e is mass of electron and m_p is mass of proton)

(A) $\sqrt{\frac{m_p}{m_e}}$

(B) $\frac{1}{2}\sqrt{\frac{m_e}{m_p}}$

(C) $\sqrt{\frac{m_e}{m_p}}$

(D) $2\sqrt{\frac{m_p}{m_e}}$

Q.4

An electron of mass ' m ' and charge ' e ' initially at rest gets accelerated by a constant electric field E. The rate of change of de-Broglie wavelength of this electron at time t, ignoring relativistic effects is :

(A) $\frac{-h}{eEt^2}$

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(B) $\frac{-eht}{E}$

(C) $\frac{-mh}{eEt^2}$

(D) $\frac{-h}{eE}$

Q.5

A proton and a deuteron are accelerated through the same accelerating potential. Which one of the two has greater value of de-Broglie wavelength associated with it, and less momentum?

(A) Proton

(B) Deutron

(C) Both have equal values

(D) None of these

Q.6

A particle of mass M at rest decays into two particles of masses m_1 and m_2, having non-zero velocities. The ratio of the de Broglie wavelengths of the particles, λ_1/λ_2, is

(A) m_1/m_2

(B) m_2/m_1

(C) 1.0

(D) $\sqrt{m_2}/\sqrt{m_1}$

Q.7

An electron microscope uses electrons accelerated by a voltage of $50\mathrm{kV}$. If λ be the deBroglie wavelength associated with the electrons. Also taking other factors, such as numerical aperture etc. to be same, the resolving power of an electron microscope be Re & that of an optical microscope which used yellow light is $\mathrm{R_o}$, then

(A) $\lambda = 0.05\tilde{A}\ldots, \mathrm{R_e} \gg \mathrm{R_o}$

(B) $\lambda = 0.15\tilde{A}\ldots, \mathrm{R_e} << \mathrm{R_o}$

(C) $\lambda = 0.11\tilde{A}\ldots, \mathrm{R_e} < \mathrm{R_o}$

(D) None of these

Q.8

Electrons with de-Broglie wavelength λ fall on the target in an X-ray tube. The cut-off wavelength of the emitted X-rays is

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(A) $\lambda_0 = \frac{2mc\lambda^2}{h}$

(B) $\lambda_0 = \frac{2h}{mc}$

(C) $\lambda_0 = \frac{2m^2c^2\lambda^3}{h^2}$

(D) $\lambda_0 = \lambda$

Q.9

The de-Broglie wavelength of a neutron in thermal equilibrium with heavy water at a temperature T (Kelvin) and mass m, is :-

(A) $\frac{\mathrm{h}}{\sqrt{3\mathrm{mkT}}}$

(B) $\frac{2h}{\sqrt{3mkT}}$

(C) $\frac{2\,\mathrm{h}}{\sqrt{\mathrm{mkT}}}$

(D) $\frac{\mathrm{h}}{\sqrt{\mathrm{mkT}}}$

Q.10

A homogeneous ball (mass $=\mathrm{m}$) of ideal black material at rest is illuminated with a radiation having a set of photons (wavelength $=\lambda$), each with the same momentum and the same energy. The rate at which photons fall on the ball is n. The linear acceleration of the ball is

(A) $\mathrm{m}\lambda\mathrm{nh}$

(B) $\mathrm{nh}/\mathrm{m}\lambda$

(C) $2\mathrm{nh}/\mathrm{m}\lambda$

(D) $2\,\mathrm{m}\lambda\mathrm{nh}$

Q.11

A particle of mass m is projected from ground with velocity u making angle θ with the vertical. The deBroglie wavelength of the particle at the highest
point is $-$

(A) ∞

(B) $\frac{\mathrm{h}}{\mathrm{mu}\sin\theta}$

(C) $\frac{\mathrm{h}}{\mathrm{mu}\cos\theta}$

(D) $\frac{\mathrm{h}}{\mathrm{mu}}$

Q.12

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An electron of mass m and a photon have same energy E. The ratio of de-Broglie wavelengths associated with them is :

(A) $\frac{1}{c}\left(\frac{E}{2m}\right)^{\frac{1}{2}}$

(B) $\left(\frac{\mathrm{E}}{2\,\mathrm{m}}\right)^{\frac{1}{2}}$

(C) $\mathrm{c}(2\mathrm{mE})^{\frac{1}{2}}$

(D) $\frac{1}{\mathrm{xc}}\left(\frac{2\,\mathrm{m}}{\mathrm{E}}\right)^{\frac{1}{2}}$

Q.13

The maximum kinetic energy of the electrons hitting a target so as to produce X -ray of wavelength $1\tilde{A}\ldots$ is

(A) $1.24\mathrm{keV}$

(B) $12.4\mathrm{keV}$

(C) $124\mathrm{keV}$

(D) None of these

Q.14

When the X-ray tube is operated at $1\mathrm{kV}$, then X-rays of minimum wavelength $6.22\tilde{A}\ldots$ are produced. If the tube is operated at $10\mathrm{kV}$, then the minimum wavelength of x -rays will be

(A) $0.622\tilde{A}\ldots$

(B) $6.22\tilde{A}\ldots$

(C) $3.11\tilde{A}\ldots$

(D) zero

Q.15

X-rays are produced in an X -ray tube operating at a given accelerating voltage. The wavelength of the continuous X-rays has values from

(A) 0 to ∞

(B) λ_{min} to ∞ where $\lambda_{\text{min}} > 0$

(C) 0 to λ_{max} where $\lambda_{\text{max}} < \infty$

(D) λ_{min} to λ_{max} where $0 < \lambda_{\text{min}} < \lambda_{\text{max}} < \infty$

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Q.16

The ratio of energies of X-rays of the wavelength $0.01\tilde{A}\ldots$ and $0.5\tilde{A}\ldots$ will be

(A) $1:1$

(B) $1:2$

(C) $1:5$

(D) $50:1$

Q.17

An x -ray tube is operating at $30\mathrm{KV}$ then the minimum wavelength of the x -rays coming out of the tube is $-$

(A) $1.24\tilde{A}\ldots$

(B) $0.413\tilde{A}\ldots$

(C) $0.124\tilde{A}\ldots$

(D) $0.13\tilde{A}\ldots$

Q.18

When the minimum wavelength of X -rays is $2\tilde{A}\ldots$ then the applied potential difference between cathode and anticathode will be

(A) $6.2\mathrm{kV}$

(B) $2.48\mathrm{kV}$

(C) $24.8\mathrm{kV}$

(D) $62\mathrm{kV}$

Q.19

The energy in monochromatic X-rays of wavelength $1\tilde{A}\ldots$ is roughly equal to

(A) 2×10^{-15} J

(B) 2×10^{-16} J

(C) 2×10^{-17} J

(D) 2×10^{-18} J

Q.20

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The short wavelength limit of continuous X radiation emitted by an X -ray tube operating at $30\mathrm{kV}$ is $0.414\tilde{A}$ . . .. Calculate Planck's constant.

(A) $6.22 \times 10^{-34}\mathrm{erg}-\mathrm{sec}$

(B) $6.624 \times 10^{-24}\mathrm{erg}-\mathrm{sec}$

(C) $6.624 \times 10^{-27}\ \mathrm{J}-\mathrm{sec}$

(D) $6.624 \times 10^{-34}\ \mathrm{J}-\mathrm{sec}$

Answers & Solutions

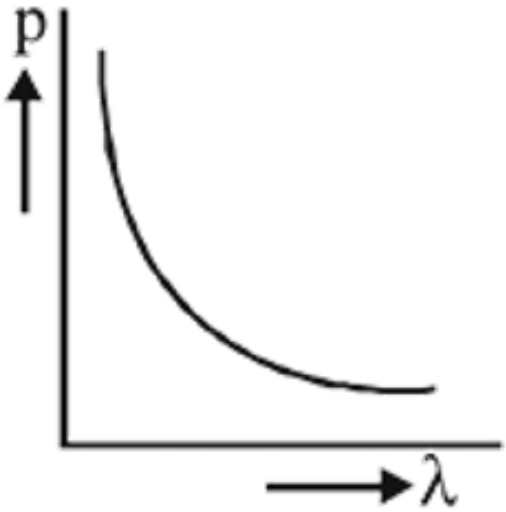
Q.1 Answer:

$200P$

Solution:

The de-Broglie's wavelength associated with the moving electron $\lambda = \frac{h}{P}$ **Now, according to problem** $\frac{d\lambda}{\lambda} = -\frac{dp}{P}; \quad \frac{0.5}{100} = \frac{P}{P'} \Rightarrow P' = 200P$

Q.2 Answer:



Solution:

According to De-broglie $\mathrm{p} = \frac{\mathrm{h}}{\lambda}$ **or** $\mathrm{P} \propto \frac{1}{\lambda}$
$\mathrm{P} \propto \frac{1}{\lambda}$ **represents rectangular hyperbola.**

Q.3 Answer:

$2\sqrt{\frac{m_p}{m_e}}$

Solution:

Energy in joule (E) **= charge** $\times$ **potential diff. in volt** $\mathrm{E_{electron}} = \mathrm{q_e V}$ **and** $\mathrm{E_{proton}} = \mathrm{q_p 4\ V}$
de-Broglie wavelength $\lambda = \frac{\mathrm{h}}{\mathrm{P}} = \frac{\mathrm{h}}{\sqrt{2\mathrm{mE}}}$
$\lambda_\mathrm{e} = \frac{\mathrm{h}}{\sqrt{2\ \mathrm{m_e eV}}}$ **and** $\lambda_\mathrm{P} = \frac{\mathrm{h}}{\sqrt{2\ \mathrm{m_P e4\ V}}}$

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$(\because q_e = q_P)$

$\therefore \frac{\lambda_e}{\lambda_P} = \frac{\frac{\sqrt{2\,m_e eV}}{h}}{\frac{\sqrt{2\,m_P e4\,V}}{=}} \sqrt{\frac{2\,m_P e4\,V}{2\,m_e eV}} = 2\sqrt{\frac{m_P}{m_e}}$

Q.4 Answer:

$\frac{-h}{eEt^2}$

Solution:

Acceleration, $a = \frac{Ee}{m}$ **Velocity of electron,** $v = at = \frac{Eet}{m}$ **Wavelength,** $\lambda = \frac{h}{mv} = \frac{h}{Eet}$.
Now $\frac{d\lambda}{dt} = -\frac{h}{eEt^2}$.

Q.5 Answer:

Proton

Solution:

de-Broglie wavelength of a charged particle
is given by, $\lambda \propto \frac{1}{\sqrt{mq}}$
If m_p **and e are mass and charge of a proton respectively, and,** m_D **and** e **are mass and charge of a deutron respectively, then** $\frac{\lambda_p}{\lambda_D} = \sqrt{\frac{m_D q_D}{m_p q_p}} = \sqrt{\frac{(2\,m_p)(e)}{(m_p)(e)}} = \sqrt{2}$

$\lambda_p = \sqrt{2}\lambda_D$

Momentum, is given by, $P = \frac{h}{\lambda}$ **or,** $p \propto \frac{1}{\lambda}$
where, $h =$ **plank's constant Since the wavelength of a proton is more than that of deutron thus, the momentum of** a **proton is lesser than that of deutron. Hence, the momentum of proton is less.**

Q.6 Answer:

1.0

Solution:

Applying conservation of linear momentum, Initial momentum = Final momentum
$0 = m_1v_1 - m_2v_2 \Rightarrow m_1v_1 = m_2v_2$
Now, $\frac{\lambda_1}{\lambda_2} = \frac{h/m_1v_1}{h/m_2v_2} = 1$

Q.7 Answer:

$\lambda = 0.05\tilde{A} \ldots, R_e \gg R_o$

Solution:

Given:- Accelerating voltage $= V = 50kV$
$= 50 \times 10^3\ V$

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Solution:-Using de-broglie's equation.

$\lambda = \frac{\mathrm{h}}{\mathrm{p}} \longrightarrow (1)$

$\lambda =$ **Wavelength of moving particle.**

$\mathrm{P} =$ **Momentum of moving particle.**

$\mathrm{h} =$ **Planck's constant** $= 6.63 \times 10^{-3} \mathrm{Js}$

now from mechanics we know,

$P = \sqrt{2mk} = \sqrt{2mqV}$

Where, $\mathrm{P} =$ **momentum of particle**

$M =$ **mass of particle.**

$\mathrm{K} =$ **Kinetic energy of particle.**

$\mathrm{V} =$ **Electric potential of particle(Accelerating voltage).**

$\mathrm{q} =$ **charge of particle.**

In the case of electron, $\mathrm{q} = \mathrm{e} = 1.6 \times 10^{-19} \mathrm{C}$

Substitution of (1)

$$\therefore \quad \lambda = \frac{\mathrm{h}}{\sqrt{2\mathrm{meV}}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 50 \times 10^{3}}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{120.66 \times 10^{-24}}$$

$$\lambda = 5.467 \times 10^{-12}\ \mathrm{m}$$

Whereas wavelength of yellow light, $\lambda_y = 590\ \mathrm{nm}$

$= 5.9 \times 10^{-7}\ \mathrm{m}$

The wavelength of the accelerated electron is about 10^5 **times less than the yellow light.**

Since resolving power of a microscope is inversely proportional to the wavelength of light hence the resolving power of the electron microscope is about 10^5 **times more than that of an optical microscope.**

Q.8 Answer:

$\lambda_0 = \frac{2mc\lambda^2}{h}$

Solution:

The cut off wavelength is given by $\lambda_0 = \frac{hc}{eV}$

According to de Broglie equation $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$

$\Rightarrow \lambda^2 = \frac{h^2}{2meV} \Rightarrow V = \frac{h^2}{2me\lambda^2}$

From (i) and (ii), $\lambda_0 = \frac{hc \times 2me\lambda^2}{eh^2} = \frac{2mc\lambda^2}{h}$

Q.9 Answer:

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$\frac{h}{\sqrt{3mkT}}$

Solution:

We know that, de-Broglie wavelength $\lambda = \frac{\hbar}{P} = \frac{h}{\sqrt{2\,m(KE)}}$

K.E. of thermal neutron $= \frac{3}{2}kT$

$= \frac{h}{\sqrt{2\,m\left(\frac{3}{2}kT\right)}} \Rightarrow \lambda = \frac{h}{\sqrt{3mkT}}$

Q.10 Answer:

$nh/m\lambda$

Solution:

Momentum imparted per unit time $= np \Rightarrow \quad F = \frac{nh}{\lambda} \quad \therefore$ **Acceleration** $= \frac{nh}{m\lambda}$

Q.11 Answer:

$\frac{h}{mu\sin\theta}$

Solution:

Velocity at highest point $= u\sin\theta . \therefore \lambda_D = \frac{h}{mu\sin\theta}$ **(Since** θ **is velocity wrt vertical)**

Q.12 Answer:

$\frac{1}{c}\left(\frac{E}{2m}\right)^{\frac{1}{2}}$

Solution:

For electron De-Broglie wavelength,

$\lambda_e = \frac{h}{\sqrt{2mE}}$; **For photon** $E = pc$

$\Rightarrow$ **De-Broglie wavelength,** $\lambda_{Ph} = \frac{hc}{E}$

$\therefore \quad \frac{\lambda_e}{\lambda_{Ph}} = \frac{h}{\sqrt{2mE}} \times \frac{E}{hc} = \left(\frac{E}{2\,m}\right)^{1/2}\frac{1}{c}$

Q.13 Answer:

$12.4keV$

Solution:

$\lambda_{min} = 1\tilde{A} \ldots$ (**given**)

$\because \lambda_{min} = \frac{1240}{E}(eV)(nm)$

Thus, $E = \frac{1240(eV)(nm)}{0.01(\,nm)} = 12400eV$

$E = 12.4KeV$

Q.14 Answer:

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$0.622\tilde{A}\ldots$

Solution:

$\frac{\lambda_{m_2}}{\lambda_{m_1}} = \frac{V_1}{V_2} \Rightarrow \lambda_{m_2} = \frac{6.22\times10^3}{10^4} = 0.622\tilde{A}\ldots$

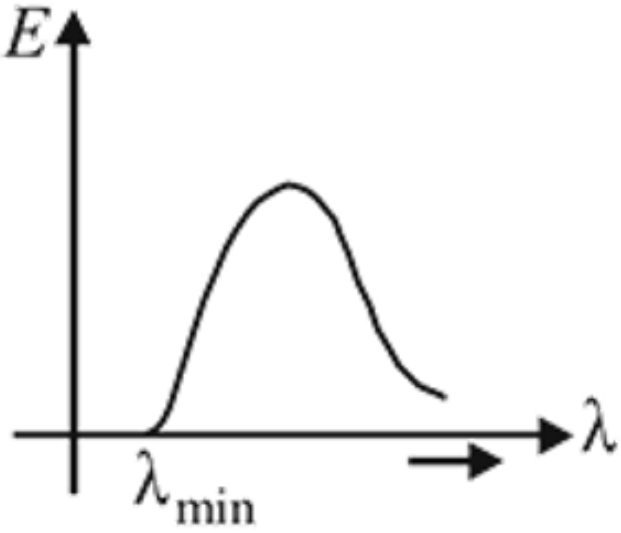
Q.15 Answer:

$\lambda_{\min}$ to ∞ where $\lambda_{\min} > 0$

Solution:

The continuous X-ray spectrum is shown in figure. All wavelengths $> \lambda_{\min}$ are found, where $\lambda_{\min} = \frac{12400}{V(\text{ in volts})}\tilde{A}\ldots$

Here V is the applied voltage.



Q.16 Answer:

$50:1$

Solution:

$\frac{E_1}{E_2} = \frac{hc}{\lambda_1}\frac{\lambda_2}{hc} = \frac{\lambda_2}{\lambda_1} \Rightarrow \frac{E_1}{E_2} = \frac{50}{1}$

Q.17 Answer:

$0.413\tilde{A}\ldots$

Solution:

$\lambda_{\min} = \frac{12400}{V}\tilde{A}\ldots = \frac{12400}{30000} = 0.413\tilde{A}\ldots$

Q.18 Answer:

6.2kV

Solution:

$V = \frac{12400}{\lambda_{\min}(\tilde{A}\ldots)}$ **Volt** $\Rightarrow V = \frac{12400}{2} = 6.2\text{KV}$

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Q.19 Answer:

2×10^{-15} J

Solution:

$$E = hc/\lambda = 6.6 \times 10^{-34} \times 3 \times 10^{8}/1 \times 10^{-10}$$
$$= 2 \times 10^{-15} \text{ J}$$

Q.20 Answer:

6.624×10^{-34} J $-$ sec

Solution:

$\lambda_{min} = \frac{ch}{eV} \therefore h = \left(\frac{eV}{c}\right)\lambda_{min}$
$h = \frac{1.6\times10^{-19}\times30000\times0.141\times10^{-10}}{3\times10^{8}}$
$\Rightarrow h = 6.624 \times 10^{-34}$ J $-$ sec

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